

Toward Stable and Clinically Reliable AI for Whole-Slide Detection of Prostate Cancer

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Abstract

Automated detection of prostatic carcinoma on whole-slide images remains challenged by variation across specimen types, institutions, laboratory work-

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flows, and model states selected during development. We developed a Stability-Prioritized Ensemble (SPE) that favored checkpoints with sustained validation performance before integrating predictions across multiple-instance learning architectures and patient-disjoint cross-validation folds. The framework was developed using 2,235 WSIs from 1,349 patients and was locked before evaluation in a temporally independent internal cohort of 345 WSIs and three external cohorts comprising 521 WSIs. SPE maintained favorable diagnostic performance across the internal and external cohorts and across core needle biopsy, transurethral resection, and radical prostatectomy specimens, achieving a balanced accuracy of 96.55% in the pooled external cohort. Compared with an ensemble retaining the peak-validation checkpoint from every training trajectory, SPE reduced the coefficient of variation of balanced accuracy by 19.50% during fixed-model resampling. Reproducibility was further evaluated in 360 matched serial-section WSIs from 60 cases processed independently at six laboratories; 96.67% of cases received concordant classifications in at least five laboratories, with generally low within-case variation in malignancy probabilities. Cancer-area-fraction analysis identified very limited tumor extent as a potential failure condition, while a descriptive reader comparison and auxiliary lesion localization provided supportive clinical context. These findings support a reliability-oriented approach to prostate WSI analysis that combines model-state consistency, independent validation, interlaboratory reproducibility, and clinically interpretable subgroup assessment.

Keywords: prostate cancer, whole-slide imaging, multiple-instance learning, ensemble learning, external validation, interlaboratory reproducibility

1. Introduction

Prostate cancer is among the most commonly diagnosed malignancies in men.[1] Histopathologic examination remains central to diagnosis and clinical management.[2, 3] Routine prostate pathology includes core needle biopsy (CNB), transurethral resection of the prostate (TURP), and radical prostatectomy (RP) specimens, each addressing a different point in the diagnostic and treatment pathway. In CNB and TURP specimens, recognition of carcinoma directly establishes or supports a diagnosis of malignancy. In RP specimens, identification of tumor-bearing sections is an initial step toward tumor quantification, pathologic staging, and margin assessment.[2, 3] Al-

11 though the downstream interpretation differs among these specimens, reli-
12 able identification of tumor-containing tissue is a common requirement. This
13 task is complicated by differences in tissue volume, architecture, tumor dis-
14 tribution, cautery, benign mimickers, staining, and scanning across routine
15 specimen types and institutions.[4, 5]

16 Weakly supervised multiple-instance learning (MIL) has enabled whole-
17 slide image (WSI) classification using routine slide-level diagnoses rather
18 than exhaustive tumor annotations.[6–8] Pathology foundation models pro-
19 vide transferable histologic representations for downstream slide analysis.[9–
20 11] These approaches have shown broad applicability across computational
21 pathology tasks, but high aggregate performance does not by itself establish
22 reliability in practice. Differences in specimen composition, tumor burden,
23 disease prevalence, tissue preparation, staining, scanning, and case mix may
24 occur simultaneously across institutions. Consequently, pooled performance
25 can obscure clinically relevant failure modes in particular specimens or set-
26 tings. Computational demands are also relevant because higher-resolution
27 processing may substantially increase analysis time without a proportional di-
28 agnostic benefit. Evaluation should therefore extend beyond overall discrimi-
29 nation to include independent validation, specimen-specific performance, pro-
30 cessing efficiency, and review of discordant cases.[12, 13]

31 A second source of variability arises during model development. In rou-
32 tine pathology, repeated analysis of the same WSI should yield reproducible
33 case-level outputs, because inconsistent predictions may undermine confi-
34 dence and limit practical use. Yet models saved at adjacent training epochs
35 may have similar validation scores while assigning different probabilities, or
36 occasionally different classifications, to the same slide. Cross-validation folds
37 expose models to different patient subsets, whereas alternative MIL architec-
38 tures summarize slide-level evidence using different aggregation mechanisms.[6,
39 8, 14, 15] Conventional ensembles often retain the single highest-scoring
40 checkpoint from each training trajectory and average all predictions equally.
41 This approach can improve mean performance, but it may also give equal
42 influence to isolated validation peaks and to model states whose predictions
43 are less consistent. These considerations motivate ensemble strategies that
44 prioritize prediction reproducibility in addition to average diagnostic perfor-
45 mance.

46 Model-side stability does not ensure that predictions remain reproducible
47 when tissue from the same case is processed through different laboratory
48 workflows. Staining and image-acquisition variation can alter tissue appear-

ance in digitized slides.[4, 5, 16] Color-normalization methods have been developed to reduce stain-related variation.[17–19] Standard multicenter validation cannot isolate these effects because technical variation is confounded by differences in patients, specimen types, and disease prevalence. Matched serial sections prepared and scanned independently across laboratories provide a more controlled assessment of whether slide-level classifications and malignancy probabilities remain reproducible under routine technical variation. Although serial sections are not microscopically identical, this design reduces, rather than eliminates, tissue-related confounding.

In this study, we developed a Stability-Prioritized Ensemble (SPE) for WSI-level detection of prostatic carcinoma across CNB, TURP, and RP specimens. SPE favored checkpoints with sustained validation performance before combining predictions across cross-validation folds and heterogeneous MIL architectures. In parallel, pathology foundation model selection, equivalent magnification, and stain normalization were evaluated sequentially to balance diagnostic performance, computational efficiency, and workflow practicality. The complete framework was locked before evaluation in a temporally independent internal cohort and three external institutional cohorts. We additionally assessed performance by specimen type and cancer area fraction and evaluated matched serial sections processed across six laboratories. A descriptive reader comparison and an auxiliary lesion-localization analysis were included to provide supportive clinical context. Together, these evaluations addressed complementary dimensions of reliability: model-state consistency, cross-specimen and cross-institutional generalizability, interlaboratory reproducibility, and computational practicality.

2. Materials and Methods

This retrospective, multi-institutional study comprised model development, a temporally independent internal test, three-center external validation, and supportive interlaboratory, reader, and localization analyses. The WSI representation, trained models, ensemble procedure, and classification threshold were locked before independent evaluation. The WSI was the primary unit of analysis; patients were kept separate during partitioning and treated as the clustering unit for statistical inference when multiple WSIs were available.

2.1. Study Design, Cohorts, and Reference Standard

The task was WSI-level detection of prostatic carcinoma in hematoxylin and eosin (H&E)-stained core needle biopsy (CNB), transurethral resection of the prostate (TURP), and radical prostatectomy (RP) specimens.[2, 3] The development cohort included 2,235 WSIs from 1,349 patients; the internal-test cohort included 345 WSIs from 138 patients; and three geographically distinct hospitals contributed 521 WSIs from 106 patients for external validation. All WSIs from one patient were assigned to the same dataset and cross-validation fold.

Eligible WSIs contained sufficient tissue for review and had a definitive slide-level benign or malignant diagnosis. Duplicate records and slides lacking a reference diagnosis were excluded. Reference labels were based on the final institutional diagnosis and senior pathologist review; diagnostically challenging cases were adjudicated using corresponding H&E and pancytokeratin/p63 immunohistochemical sections when available. The study was approved by the institutional ethics committee (K2025-12-030), conducted in accordance with the Declaration of Helsinki, and granted a waiver of informed consent for de-identified retrospective material.

Cancer area fraction was defined as the proportion of evaluable tissue occupied by carcinoma on a tumor-containing WSI and categorized as very low ($< 1\%$), low (1% to $< 5\%$), medium (5% to $< 20\%$), or high ($\geq 20\%$). Supportive datasets comprised 360 matched serial-section WSIs from 60 cases processed at six laboratories, 100 external-cohort WSIs reviewed by two pathologists, and 25 patient-disjoint tumor-containing WSIs used for localization assessment. None contributed to model development or configuration selection. Complete patient-disjoint out-of-fold (OOF) predictions were available for 2,231 of 2,235 development WSIs; the remaining four were excluded only from paired OOF analyses.

2.2. WSI Representation and Model Development

Tissue regions were identified by Otsu thresholding within the CLAM workflow, and nonoverlapping tissue patches were extracted after background exclusion [8]. Resolution was standardized using microns-per-pixel metadata. Seven pretrained pathology foundation models—UNI, UNI2, CONCH, Virchow2, mSTAR, OmiCLIP, and H-optimus-1—were compared at $20\times$ equivalent magnification using a fixed attention-based MIL classifier without stain normalization [9–11, 20–23]. After encoder selection, $5\times$, $10\times$, and $20\times$ representations were compared, followed by an evaluation of no normalization,

Reinhard, Macenko, and Vahadane normalization, while all other components were held constant [17–19]. Diagnostic performance was the primary selection criterion, with processing time considered when results were similar. Processing time included WSI loading, tissue segmentation, patch preparation, optional stain normalization, and feature extraction. H-optimus-1 at 10× with Reinhard normalization was subsequently selected and locked for all downstream experiments. Field sizes, normalization procedures, hardware specifications, and timing conditions are detailed in the Supplementary Methods.

Eleven MIL architectures were trained using identical feature bags and patient-level folds: AB-MIL, CLAM-SB, CLAM-MB, TransMIL, WIKG-MIL, Mamba2D-MIL, AEM-MIL, MICO-MIL, MSM-MIL, TDA-MIL, and GDF-MIL.[6, 8, 14, 15, 24–29] Models were trained with Adam, an initial learning rate of 2×10^{-4} , a batch size of one WSI, a maximum of 50 epochs, and early stopping after 30 epochs without improvement in validation macro-F1. Five-fold patient-disjoint cross-validation generated five trajectories per architecture. Each model returned a malignancy probability, with classification at a fixed threshold of 0.5.

2.3. *Stability-Prioritized Ensemble Construction*

For architecture a and fold k , validation macro-F1 at epoch e was denoted by $F_{a,k}(e)$. A stable interval comprised at least five consecutive epochs satisfying

$$|F_{a,k}(e) - F_{a,k}(e - 1)| \leq 0.003.$$

When several intervals qualified, the final interval was retained, and the checkpoint with the highest macro-F1 within that interval was selected. If no interval qualified, the best checkpoint among the final five available epochs was used. SPE probabilities were the equal-weight mean of the resulting 55 architecture-fold predictions.

Comparators were constructed from the same trained models. Include-All retained the peak-validation checkpoint from every trajectory; Top-5 retained the five architectures with the highest mean peak-validation macro-F1 across folds; and Random used a prespecified fixed set of five architectures. All ensembles used equal-weight probability averaging. Architecture sets, checkpoint rules, aggregation procedures, and the threshold were determined using development data and were unchanged during independent evaluation. Further implementation details are provided in the Supplementary Methods.

2.4. Independent Validation and Supportive Analyses

The locked SPE was applied without recalibration or further tuning to the internal-test and external cohorts. Performance was summarized overall and, where sample sizes permitted, by institution and specimen type. In the pooled external cohort, sensitivity, false-negative rate, and malignancy-probability distributions were evaluated within each cancer-area stratum. Specificity was calculated from the complete tumor-free external reference set and used as a common reference for stratum-specific balanced accuracy; AUROC was not calculated because each area stratum contained only tumor-containing WSIs. Fixed-model resampling of the internal-test cohort assessed sensitivity to test-cohort composition without retraining or checkpoint reselection.

For the matched interlaboratory analysis, one serial section from each of 30 tumor-containing and 30 tumor-free cases was processed using routine H&E staining and scanning at each of six laboratories. The locked pipeline was applied without laboratory-specific adaptation, and staining appearance was characterized using optical-density distributions after color deconvolution.

The reader study included 70 tumor-containing and 30 tumor-free external-cohort WSIs. Pathologists with seven and five years of diagnostic experience independently reviewed the slides in QuPath.[30] They had not previously reviewed the study slides and were blinded to clinical information, immunohistochemistry, reference labels, model outputs, and each other’s assessments. Navigation and magnification were unrestricted; submitted decisions could not be revised. Performance, agreement, interpretation time, and error overlap were summarized descriptively.

For localization, reference tumor regions were established by two pathologists who were independent of the study readers and had access to matched immunohistochemistry. A YOLO11s detector was trained and validated using 7,045 and 5,180 patches, respectively, from 16 WSIs representing 10 patients, with patient-level separation.[31, 32] Training used stochastic gradient descent for 500 epochs and 1536×1536 -pixel inputs. At inference, detections below 0.30 confidence were removed, followed by class-agnostic nonmaximum suppression at an intersection-over-union threshold of 0.40. The two readers independently delineated suspected tumor regions after classification while remaining blinded to the reference annotations and model outputs. Reader and detector masks were compared with the reference masks in 25 patient-disjoint tumor-containing WSIs using Dice coefficient and intersection over

union. Localization outputs did not contribute to SPE classification.

2.5. Statistical Analysis

Analyses were performed at the WSI level, with the patient as the clustering unit when multiple WSIs were present. Tumor-containing WSIs were positive. Accuracy, balanced accuracy, sensitivity, specificity, positive and negative predictive values, macro-F1, AUROC, and Cohen’s κ were calculated; threshold-dependent metrics used 0.5, whereas AUROC used continuous probabilities.

OOB configuration CIs used 2,000 patient-level bootstrap replicates. Configuration differences were evaluated by paired patient-level bootstrap, with two-sided P values calculated as twice the smaller proportion of differences on either side of zero. For internal and external testing, Wilson intervals were used for sensitivity and specificity, and 5,000 patient-clustered bootstrap replicates were used for balanced accuracy, AUROC, macro-F1, and κ ; pooled external resampling was additionally stratified by institution.

Ensemble variability was assessed using 1,000 patient-clustered fixed-model resamples and therefore did not capture retraining variance. Interlaboratory reproducibility was summarized by laboratory-specific performance, pairwise agreement and κ , cross-laboratory classification concordance, and within-case SD of malignancy probabilities. Reader comparisons were descriptive. Metrics requiring both classes were not calculated for single-class strata. Small-subgroup, cancer-area, pathologic-review, reader, and localization analyses were exploratory and were not adjusted for multiple comparisons. Missing variables were not imputed. All tests were two-sided, and effect estimates with 95% CIs were emphasized. Analyses used Python 3.10 with fixed random seeds.

3. Results

The primary cohorts comprised 3,101 WSIs from 1,593 patients, including the development cohort, the temporally independent internal-test cohort, and three external institutional cohorts. CNB, TURP, and RP specimens were represented. Complementary analyses included matched serial sections processed across six pathology laboratories, a reader-study cohort, and a localization subset enriched for minute or morphologically subtle cancer foci. Study design, cohort assembly, and clinicopathologic characteristics are summarized in Fig. 1 and Supplementary Table S1.

228 *3.1. Selection of a Computationally Efficient WSI Representation*

229 Patient-disjoint OOF comparisons identified H-optimus-1 as the preferred
230 feature encoder. The leading foundation models performed similarly, and the
231 balanced-accuracy difference between H-optimus-1 and the strongest alter-
232 native was small. Encoder selection therefore reflected the overall diagnostic
233 profile rather than a large advantage of a single model (Fig. 2A; Supplemen-
234 tary Tables S2–S3).

235 Reducing the equivalent magnification from $20\times$ to $10\times$ shortened mean
236 end-to-end preprocessing and feature-extraction time by approximately three-
237 fold, with a modest decrease in diagnostic performance. Processing at $5\times$
238 was faster but produced a larger reduction in performance. These findings
239 supported $10\times$ as a pragmatic operating point that balanced preservation of
240 diagnostic information with WSI-processing throughput, rather than estab-
241 lishing equivalence across magnifications (Fig. 2B).

242 At $10\times$, Reinhard normalization improved diagnostic performance with
243 little additional processing time. Macenko and Vahadane normalization re-
244 quired substantially longer processing without a commensurate diagnostic
245 benefit. H-optimus-1 at $10\times$ with Reinhard normalization was therefore se-
246 lected as the locked representation because it provided a favorable balance
247 among diagnostic performance, computational efficiency, and workflow prac-
248 ticality (Fig. 2C–D; Supplementary Tables S2–S3).

249 *3.2. Diagnostic Performance and Variability of the Stability-Prioritized En-*
250 *semble*

251 Using the locked WSI representation, the prespecified checkpoint-selection
252 rule was applied to all training trajectories from 11 MIL architectures and
253 five patient-disjoint folds. One checkpoint was retained from the final quali-
254 fying stable interval of each trajectory, and predictions from the resulting 55
255 models were averaged to form the locked SPE.

256 In the internal-test cohort, SPE had the highest point estimates for most
257 threshold-dependent measures among the evaluated individual architectures
258 and ensemble comparators while retaining high discrimination (Fig. 3A; Sup-
259 plementary Table S4). The principal difference was observed in the balance
260 between sensitivity and specificity rather than in AUROC alone. Because
261 paired confidence intervals were unavailable for every comparison, these find-
262 ings indicate a favorable comparative profile but do not establish statistical
263 superiority over all alternatives.

264 Across 1,000 fixed-model resamples of the internal-test cohort, SPE re-
 265 tained similar mean performance and showed lower relative dispersion than
 266 Random, Top-5, and Include-All. Variability in balanced accuracy was lower
 267 than that of Include-All (Fig. 3C). Because the models and checkpoints
 268 were not retrained or reselected, this analysis reflects reduced sensitivity to
 269 changes in test-cohort composition rather than direct evidence of robustness
 270 to repeated training.

271 Performance was maintained across CNB, RP, and TURP specimens,
 272 although TURP had the lowest point estimate and the smallest sample
 273 size (Fig. 3D; Supplementary Table S5). This finding identifies TURP as
 274 a subgroup requiring further evaluation rather than establishing a specimen-
 275 specific failure.

276 *3.3. Independent Validation Across Institutions and Specimen Types*

277 The three external cohorts differed in specimen composition, tumor preva-
 278 lence, and staining characteristics. Rather than harmonizing these real-world
 279 differences, the evaluation preserved them as a clinically heterogeneous test
 280 of model transfer. The locked SPE was applied without recalibration or fur-
 281 ther tuning and maintained high diagnostic performance in the temporally
 282 independent internal-test cohort and the pooled external cohort, with no
 283 overall loss of discrimination across the three external institutions (Fig. 3B,
 284 E, and F; Supplementary Tables S4 and S6).

285 Performance varied moderately among institutions but remained favor-
 286 able in each external cohort. No institution showed a consistent failure affect-
 287 ing both tumor-containing and tumor-free WSIs. CNB and RP performance
 288 was relatively stable across analyses, whereas TURP again had the lowest
 289 point estimate. The repeated pattern suggests that TURP may be a more
 290 challenging specimen context, although several institution-specimen strata
 291 were small or contained only one reference class and were therefore inter-
 292 preted descriptively (Fig. 3E–F; Supplementary Table S7).

293 Among tumor-containing WSIs in the pooled external cohort, sensitiv-
 294 ity was lowest in the very-low cancer-area-fraction group ($< 1\%$; 88.14%),
 295 compared with 97.44%, 100.00%, and 97.44% in the low, medium, and high
 296 groups, respectively. The corresponding false-negative rates were 11.86%,
 297 2.56%, 0.00%, and 2.56%. Median malignancy probabilities remained high
 298 across strata (0.9748–0.9991), although the lower quartile was lower in the
 299 very-low group (0.7902), indicating greater dispersion in confidence when
 300 carcinoma occupied less than 1% of evaluable tissue. Using the common

external tumor-free reference set, balanced accuracy was 92.50%, 97.15%, 98.43%, and 97.15%, respectively. These descriptive findings identify very limited tumor extent as a potential failure condition but do not establish a monotonic area–performance relationship (Supplementary Table S8).

3.4. Reproducibility Across Laboratory Workflows

Color-deconvolution analysis confirmed measurable differences in hematoxylin and eosin appearance among the six laboratory workflows (Fig. 4A). The matched serial-section design retained routine interlaboratory variation while reducing, rather than eliminating, tissue-related confounding. The experiment therefore assessed the complete sectioning, staining, scanning, and prediction pathway rather than visually standardized replicate slides.

Despite these differences, diagnostic performance remained high across all six laboratories. Most cases received the same binary classification in at least five laboratories, and pairwise agreement was consistently strong (Fig. 4B; Supplementary Table S9). Thus, most slide-level decisions were preserved across the evaluated workflows, although complete concordance was not achieved for every case.

Within-case malignancy probabilities showed low overall variability, with greater variation among tumor-free than tumor-containing cases (Fig. 4C). The high overall reproducibility was therefore driven in part by consistently high-confidence predictions in tumor-containing cases, whereas benign cases contributed more to probability variation.

The few threshold-crossing cases had probabilities close to 0.5 and could also differ in the amount of diagnostically relevant tissue represented in individual serial sections. Disagreement could therefore not be attributed solely to staining or scanner effects. These findings support reproducibility across the evaluated laboratory workflows but do not establish invariance to any single technical factor (Fig. 4D).

3.5. Supportive Reader Comparison and Lesion Localization

In the reader-study cohort, SPE performed within the range observed for the two participating pathologists. Its performance was close to that of the pathologist with seven years of diagnostic experience and its balanced accuracy was higher than that of the pathologist with five years of experience. The difference between readers was driven mainly by lower sensitivity in the less experienced reader, whereas SPE showed a more balanced sensitivity–specificity profile (Fig. 5A–D; Supplementary Tables S10–S11).

337 The reader comparison was not designed to establish pathologist-level
338 equivalence. Only two readers participated, and the cohort contained a
339 higher proportion of tumor-containing WSIs than would be expected in many
340 routine settings. Interpretation time was reported to provide workflow con-
341 text but was not compared directly with computational processing time be-
342 cause human review and model inference represented different components
343 of the diagnostic process.

344 Beyond slide-level classification, the localization analysis examined whether
345 suspicious regions could be displayed to support pathologic review. In the se-
346 lected subset, the auxiliary YOLO module showed close spatial overlap with
347 pathologist-defined reference regions and numerically higher overlap than ei-
348 ther participating reader. The subset was small and deliberately enriched
349 for minute or morphologically subtle cancer foci, and no formal human–AI
350 superiority analysis was prespecified. Localization was therefore interpreted
351 as supportive spatial evidence rather than as a standalone lesion-detection
352 endpoint. It did not contribute to the SPE malignancy probability (Fig. 5E;
353 Supplementary Table S12).

354 4. Discussion

355 In this multi-institutional study, we developed SPE for WSI-level detec-
356 tion of prostatic carcinoma across CNB, TURP, and RP specimens. The
357 study was designed around sources of variability that are relevant to routine
358 pathology: model selection during development, differences in specimen type
359 and institutional case mix, technical variation across laboratory workflows,
360 and computational requirements for WSI processing. SPE showed favorable
361 diagnostic performance relative to the evaluated individual MIL architectures
362 and ensemble comparators, maintained high discrimination in three external
363 institutions, and produced largely concordant predictions across six labora-
364 tory workflows. The principal contribution is therefore not a large increase in
365 pooled AUROC, but an integrated evaluation of complementary dimensions
366 of reliability: model-state consistency, diagnostic performance across spec-
367 imens and institutions, interlaboratory reproducibility, and computational
368 practicality.

369 SPE addressed a practical limitation of selecting models solely at the
370 highest validation score. Adjacent checkpoints can have similar aggregate
371 performance while assigning different probabilities to individual WSIs. In
372 a diagnostic setting, such variability is most consequential for cases near

373 the classification threshold, where small changes in probability may alter
374 the reported result. SPE restricted checkpoint selection to periods of sus-
375 tained validation performance and then combined predictions across folds and
376 MIL architectures without additional weighting or a separately trained meta-
377 model. In fixed-model resampling, SPE showed lower performance dispersion
378 than Random, Top-5, and Include-All ensembles. This result indicates less
379 sensitivity to test-cohort composition after the ensemble had been locked. It
380 does not, however, demonstrate stability across independent retraining or all
381 plausible checkpoint selections; those questions require repeated training or
382 explicit reconstruction of alternative model states.

383 The cross-specimen analysis supports a shared but deliberately bounded
384 clinical task. In CNB and TURP specimens, slide-level tumor detection can
385 support identification of malignancy. In RP specimens, the same output
386 is more appropriately interpreted as identification of tumor-bearing sections
387 that require subsequent quantification, grading, staging, and margin assess-
388 ment. The model should therefore not be viewed as replacing specimen-
389 specific pathologic interpretation. Performance was favorable across all three
390 specimen types, although TURP had the lowest point estimates in both inter-
391 nal and external analyses. Tissue fragmentation, cautery, and heterogeneous
392 glandular morphology are plausible contributors, but the available subgroup
393 data do not establish the cause. The cancer-area-fraction analysis provided a
394 complementary stress test: sensitivity was lowest when carcinoma occupied
395 less than 1% of evaluable tissue, and this stratum showed greater lower-tail
396 dispersion in malignancy probability. This finding supports limited tumor
397 extent as a potential failure condition, but the analysis was exploratory, did
398 not include Grade Group stratification, and was not designed to establish a
399 continuous dose–response relationship.

400 Beyond ensemble construction, optimization of the WSI representation
401 also improved computational practicality. Processing at 10× reduced mean
402 WSI analysis time by approximately threefold relative to 20×, while Rein-
403 hard normalization recovered much of the associated performance differ-
404 ence with little additional computational burden. The selected configura-
405 tion should therefore be understood as a performance–efficiency compromise
406 rather than proof that lower-resolution analysis is diagnostically equivalent
407 to 20× review. In a future assisted workflow, processing time would need
408 to be evaluated together with slide-scanning time, data transfer, case vol-
409 ume, hardware availability, and the manner in which results are presented to
410 pathologists.

411 The matched interlaboratory experiment extended conventional external
412 validation by reducing case-mix confounding while retaining routine variation
413 in sectioning, staining, and scanning. Most cases retained the same classifica-
414 tion across laboratories, and within-case probability variation was generally
415 low. Variability was greater among tumor-free cases, suggesting that benign
416 or borderline-appearing patterns may contribute disproportionately to tech-
417 nical sensitivity. Serial sections are not histologically identical, however, and
418 the few threshold-crossing cases could reflect differences in represented tissue
419 as well as staining or scanner effects. The experiment therefore supports re-
420 producibility of the complete laboratory-to-prediction pathway across the six
421 evaluated sites, but it cannot isolate stain invariance or scanner invariance.

422 The reader and localization analyses provide limited but clinically inter-
423 pretable context. SPE performed within the range of the two participating
424 pathologists, and the auxiliary localization module produced spatial outputs
425 corresponding to reference tumor regions. Misclassifications were partially
426 shared but not identical among SPE and the readers, indicating that they
427 did not fail on precisely the same cases; this descriptive observation does
428 not establish incremental diagnostic benefit. These findings do not estab-
429 lish pathologist equivalence, superiority, autonomous diagnosis, or improved
430 workflow efficiency. Instead, they support further evaluation of the frame-
431 work as a decision-support system in which slide-level predictions are paired
432 with spatial guidance for triage, focused review, or quality assurance. Such
433 use cases require multi-reader studies in which pathologists review cases with
434 and without model assistance and in which diagnostic accuracy, review time,
435 error correction, reader confidence, and human–AI interaction are measured
436 directly.

437 Several limitations should be considered. The study was retrospective
438 and geographically restricted, and some institution–specimen and cancer-
439 area-fraction strata were small or imbalanced. The cancer-area analysis was
440 restricted to the pooled external cohort, used a shared tumor-free reference
441 set for balanced accuracy, and did not include Grade Group stratification.
442 The primary task was limited to WSI-level tumor detection and did not in-
443 clude grading, clinical tumor quantification, staging, margin assessment, or
444 comprehensive reporting. The checkpoint-selection rule, architecture set, en-
445 semble composition, and fixed threshold were derived from one development
446 population, and fixed-model resampling did not capture variability from in-
447 dependent retraining. The six-laboratory experiment could not fully separate
448 serial-section differences from staining, preparation, and scanner effects. The

449 reader study included only two pathologists and was not designed as an AI-
450 assisted, superiority, non-inferiority, or equivalence study, and the localiza-
451 tion analysis used a small enriched subset. Prospective assessment in routine
452 pathology workflows is needed to clarify effects on diagnostic efficiency, case
453 triage, reader confidence, and human–AI interaction. Broader evaluation
454 across populations, scanner platforms, and laboratory practices, together
455 with repeated training and extensions to Grade Group, tumor quantifica-
456 tion, and margin assessment, will be required before clinical implementation.
457 Within these limitations, the findings support evaluating WSI-based diagnos-
458 tic systems through a combination of independent diagnostic performance,
459 model-state consistency, interlaboratory reproducibility, and computational
460 practicality.

461 **Data and Code Availability**

462 The de-identified human prostate whole-slide images (WSIs) and associ-
463 ated clinicopathologic data are not publicly available because their use is re-
464 stricted by institutional data-protection requirements, ethics approvals, and
465 multicenter data-sharing agreements. De-identified derived data and pro-
466 cessed feature representations supporting the findings of this study may be
467 made available by the corresponding author upon reasonable request, sub-
468 ject to approval by the participating institutions and compliance with ap-
469 plicable ethical and legal requirements. The source code used for WSI pre-
470 processing, model training, Stability-Prioritized Ensemble (SPE) construc-
471 tion, statistical analysis, and region-level localization is publicly available
472 at <https://github.com/linboliao/MXZY-AI.git>. Additional implemen-
473 tation details, configuration files, and computational parameters required to
474 reproduce the reported analyses are provided in the Supplementary Methods
475 or are available from the corresponding author upon reasonable request.

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487 **Author Contributions**

488 Conceptualization: L.L., S.C., and L.W.; Methodology: L.L. and K.H.;
489 Software: L.L., K.H., X.W., and W.L.; Formal analysis: L.L. and S.C.; Data
490 curation: S.C., B.C., Q.Y., Z.S., Y.H., H.H., X.Q., L.Z., C.H., D.L., C.Y.,
491 and G.C.; Validation: S.C., B.C., and C.H.; Visualization: L.L. and K.H.;
492 Writing–original draft: L.L.; Writing–review and editing: L.L., S.C., K.H.,
493 X.W., W.L., C.Y., G.C., and L.W.; Supervision: G.C. and L.W. All authors
494 reviewed and approved the final manuscript.

495 **Declaration of Competing Interest**

496 S.C., B.C., and Q.Y. are employees of Fuzhou Maixin Biotech Co., Ltd.,
497 Fuzhou, China. The remaining authors declare that they have no compet-
498 ing financial interests or personal relationships that could have appeared to
499 influence the work reported in this paper.

500 **Ethics Approval and Consent to Participate**

501 The study was approved by the institutional ethics committee (approval
502 no. K2025-12-030) and was conducted in accordance with the Declaration
503 of Helsinki. The requirement for informed consent was waived because the
504 study used de-identified retrospective materials.

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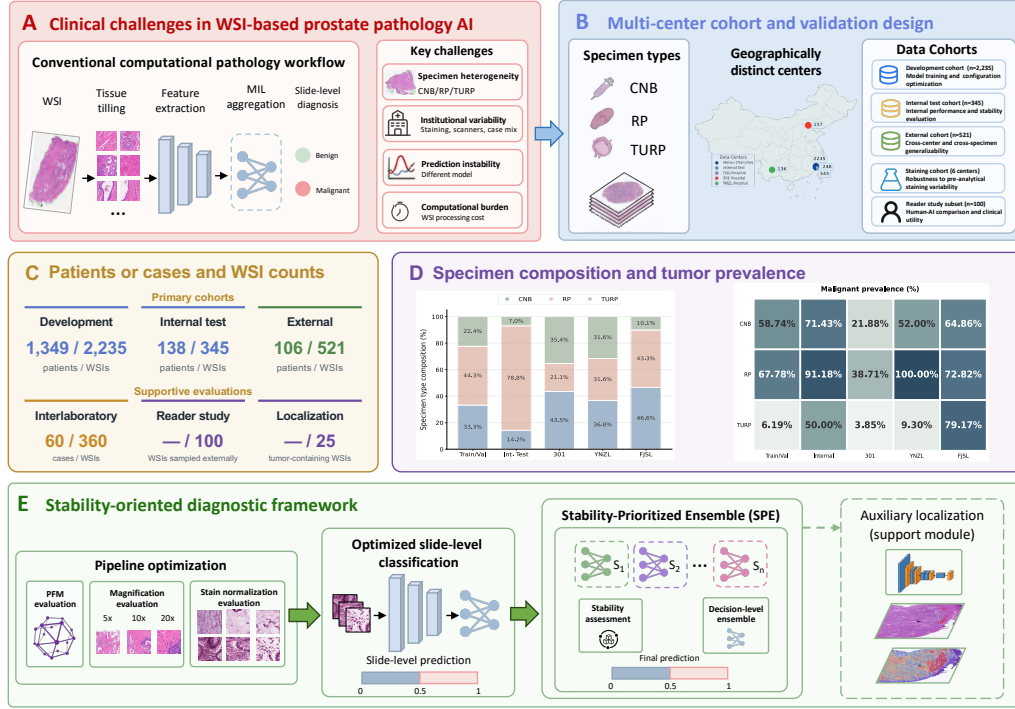


Figure 1: **Study design and stability-oriented diagnostic framework.** (A) Clinical and technical challenges in WSI-based prostate cancer detection. (B) Multicenter cohorts and validation design. (C) Patient, case, and WSI counts. (D) Specimen composition and tumor prevalence. (E) Pipeline optimization, MIL-based classification, SPE construction, and auxiliary lesion localization. CNB, core needle biopsy; MIL, multiple-instance learning; RP, radical prostatectomy; SPE, Stability-Prioritized Ensemble; TURP, transurethral resection of the prostate; WSI, whole-slide image.

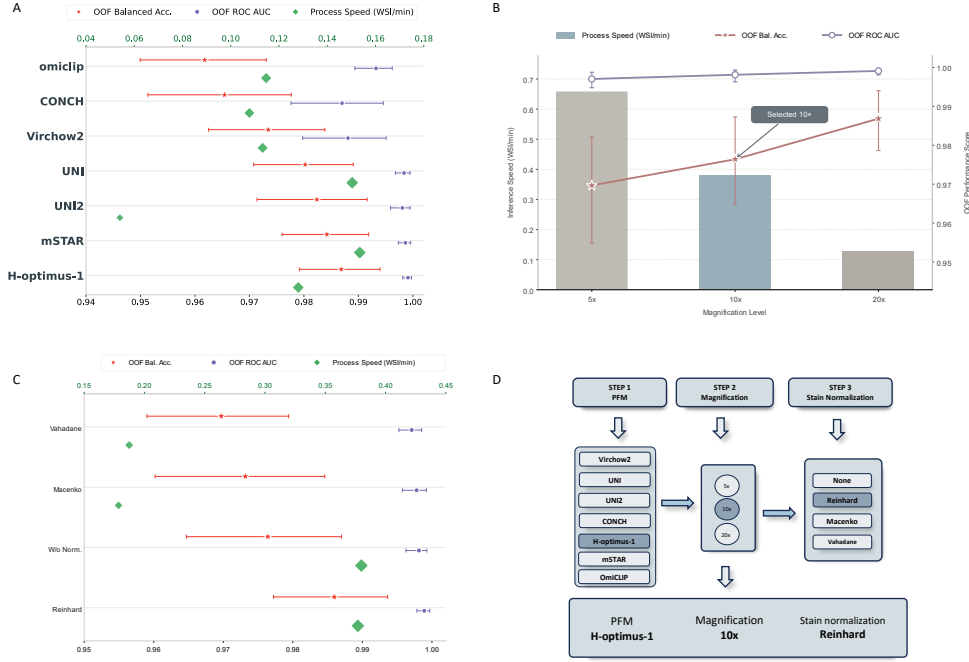


Figure 2: Development-stage selection of the diagnostic configuration. (A) OOF balanced accuracy, OOF AUROC, and processing speed for seven pathology foundation models under a fixed 20× AB-MIL setting without stain normalization. (B) OOF performance and processing speed at 5×, 10×, and 20× using H-optimus-1. Error bars are patient-bootstrap 95% CIs; 10× was selected as the performance-efficiency operating point. (C) Comparison of stain-processing strategies at 10×. Paired patient-bootstrap inference was restricted to configurations with retained aligned OOF predictions. (D) Locked configuration. AB-MIL, attention-based multiple-instance learning; AUROC, area under the receiver operating characteristic curve; CI, confidence interval; OOF, out-of-fold; WSI, whole-slide image.



Figure 3: **Diagnostic performance and subgroup analyses of the Stability-Prioritized Ensemble.** (A) Internal-test performance of SPE and individual MIL models. (B) Corresponding performance in the pooled external-validation cohort. (C) Comparison of ensemble strategies across 1,000 fixed-model patient-clustered resamples; points and error bars indicate mean and SD. (D) Internal-test performance by specimen type. (E) External-validation performance by institution. (F) External-validation performance by specimen type. (G) External-validation performance in tumor-positive WSIs by tumor area fraction. AUROC, area under the receiver operating characteristic curve; CNB, core needle biopsy; MIL, multiple-instance learning; RP, radical prostatectomy; SPE, Stability-Prioritized Ensemble; TURP, transurethral resection of the prostate.

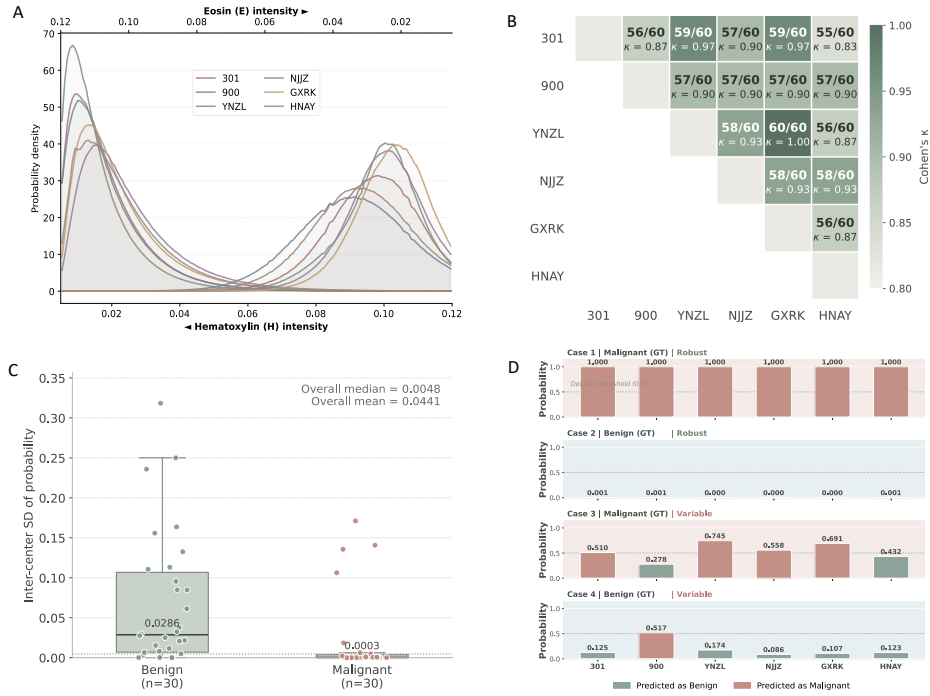


Figure 4: **Interlaboratory reproducibility across matched serial sections.** (A) Laboratory-specific hematoxylin and eosin optical-density distributions after color deconvolution. (B) Pairwise classification agreement and Cohen's kappa for 60 matched cases. (C) Within-case SD of malignancy probabilities in tumor-free and tumor-containing cases. (D) Probabilities across laboratories for representative reproducible and variable cases; the dashed line is the fixed 0.5 threshold. Serial sections reduce but do not eliminate tissue-level differences. SD, standard deviation.

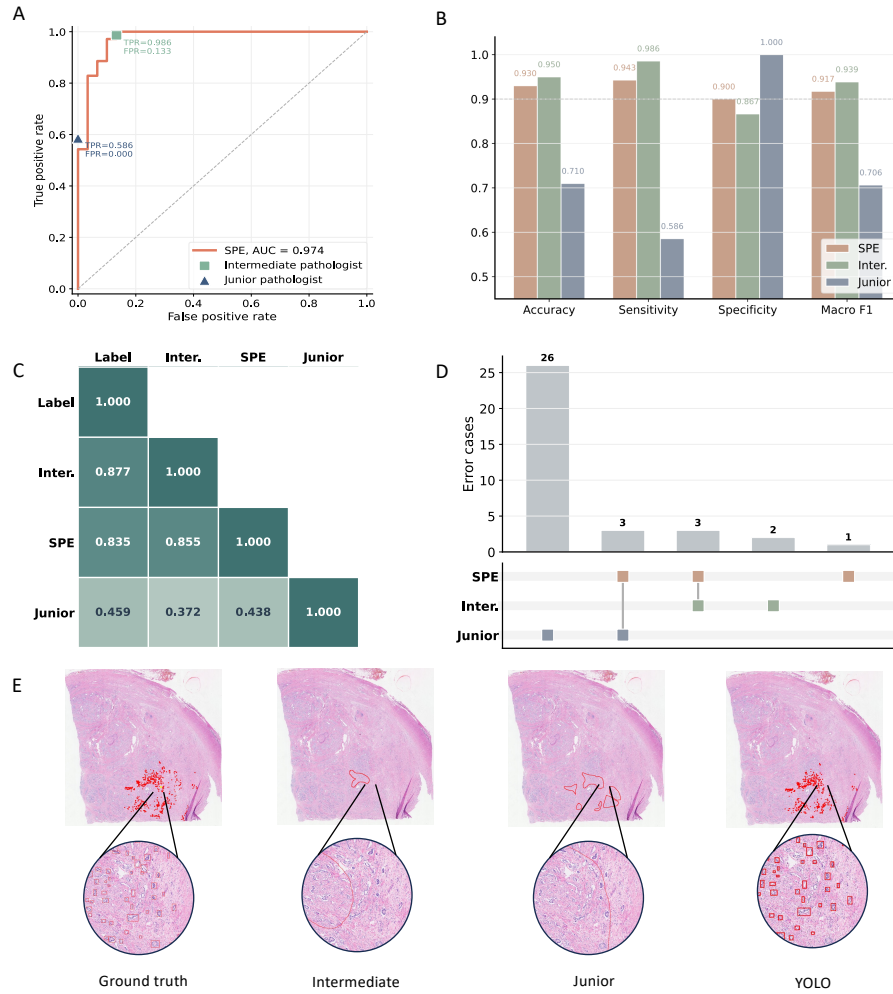


Figure 5: Reader comparison and supportive lesion localization. (A) Receiver operating characteristic curve for SPE in the 100-WSI reader subset with reader operating points. (B) Diagnostic metrics for SPE and two pathologists. (C) Pairwise Cohen's kappa. (D) Unique and shared misclassifications. (E) Representative lesion localization of a minute cancer focus by the readers and auxiliary YOLO module relative to the reference annotation. This descriptive study was not designed to establish human-AI superiority or equivalence. SPE, Stability-Prioritized Ensemble; WSI, whole-slide image; YOLO, You Only Look Once.